5G NR Network Configuration Report

1. Data Sources

The app uses real-time geospatial data from **OpenStreetMap (OSM)** via the **Overpass API**.

- **Road Networks**: Extracted to calculate the **Average Area Speed**.
- **Building Data**: Extracted to determine **User/Structure Density**.

2. Configuration Logic

The system applies the following rules to determine the optimal 5G NR parameters:

A. Sub-Carrier Spacing (SCS) vs. Speed

SCS is adjusted to combat Doppler shift caused by mobility.

- **120 kHz**: Selected if Average Speed **> 90 km/h**.
- **60 kHz**: Selected if Average Speed **> 50 km/h**.
- **30 kHz**: Selected if Average Speed **≤ 50 km/h**.

B. Frequency Band vs. Density

Bands are selected to balance capacity and coverage based on the (N) number of structures.

- **24 GHz (mmWave)**: Selected for High Density (**N > 150**). Provides maximum capacity for "City" scenarios.
- **3.5 GHz (C-Band)**: Selected for Medium Density (**50 < N <= 150**). Balanced for "Village" scenarios.
- **700 MHz (Low Band)**: Selected for Low Density (**N <= 50**). Maximum coverage for "Wilderness" scenarios.

C. Cyclic Prefix (CP) Mode

CP mode is determined by the propagation environment to handle multipath delay.

- **Normal Mode**: Used in **Dense/Urban** areas where cell sizes are small and delay spread is low.
- **Extended Mode**: Used in **Rural/Sparse** areas to protect against long-distance signal echoes in large cells.